

Python Q&A; Set 3 – Collections

Questions with clear answers for learning and interview preparation

1. What is a list?

Answer: A list is an ordered, mutable collection that can contain duplicate values.

2. What is a tuple?

Answer: A tuple is an ordered, immutable collection.

3. What is a set?

Answer: A set is an unordered collection of unique elements.

4. What is a dictionary?

Answer: A dictionary stores data as key-value pairs.

5. How can you remove duplicates while preserving order?

Answer: Use a set to track seen values while building a new list.

6. What is slicing?

Answer: Slicing extracts part of a sequence using start:stop:step.

7. `append()` vs `extend()`?

Answer: `append()` adds one object as one element; `extend()` adds elements from an iterable.

8. What does `dict.get()` do?

Answer: It returns a key's value and can return a default instead of raising `KeyError`.

9. What is a list comprehension?

Answer: It is a compact way to create a list from an iterable, optionally with a condition.

10. List vs set?

Answer: Lists preserve order and allow duplicates; sets contain unique elements and are optimized for membership operations.